

## Governance “of, with and by” AI: a European perspective on the use and impact of AI in the public sector

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*In this module, Gianluca Misuraca explores the governance “of, with and by” AI, focusing on a European perspective on the use and impact of AI in the public sector. Misuraca divides his lecture into four parts: (1) Shaping Digital Europe 2040; (2) the trilemma of the Governance “of, with and by AI”; (3) AI in public services in the EU; and the EU approach to human-centric AI.*

### Lecture Transcript

**0:00** Hello, everyone, I'm Gianluca Misuraca, and I'm the Executive Director of AI4GOV, the master on artificial intelligence for public services, that is EU co funded initiative led by the Universidad Politécnica de Madrid, the Department of Artificial Intelligence and Politecnico di Milano, Department of Design. I'll tell you more about these later on. But also, I am the Co-Founder and Vice President on Technology Diplomacy of Inspiring Futures. It is a global consultancy on strategic planning and policy support, mainly on digital governance and foresight headquartered in Switzerland and in Spain, where I'm actually based as I've been working for several years for the Joint Research Center of the European Commission that I will also mention, as part of the result of the research I've been leading on E-governance and Public Sector Innovation for several years in Seville. As part of the assignment I'm currently leading, I'm also the team leader and key expert on technology, diplomacy, and strategic communication of the InTouchAI.eu initiative that is funded by the European External Action Service for foreign policy instruments of the European Commission, and implemented in collaboration with DG Connect to promote the human centric approach on artificial intelligence of the European Union. So that I will mention later on today. As part of this assignment, I'm of course supporting the commission and the EU institution on promoting these globally. And indeed, very happy to be here. And thanks to the organizers for the invitation to provide this lecture to provide our perspective, my perspective, on the use and impact of artificial intelligence in the public sector, or what I call the governance “of, with and by” artificial intelligence.

**2:12** Clearly, AI ethics is at the core of this discussion today, in Europe and worldwide. As we will see, I tried to do that in the next half an hour, 40 minutes, by touching the four main topics: one, as you know, a futurist, I will start from the future. And so presenting results of research I've been leading on shaping Digital Europe and Horizon 2040. That was back in 2020, where we were anticipating some of the key issues that are currently on the policy and research debate. This includes the trilemma of the governance “of, with and by” AI that I mentioned before, and we will dive deeper into this later. A specific focus will be of course, on AI in public services with a specific perspective of the EU, where we are still unfortunately, navigating between rhetoric and reality. And then

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we go back to the future and I will present the EU approach to human centric AI exactly how it is embodied in the initiative I have the pleasure to lead for the European Union InTouchAI.eu.

**3:27** Now let's look at the horizon 2040. What could be Europe 20 years from the time we have been defining these scenarios? And of course, already a few years ago, and now still, there is a quite a rhetorical, I would say, claim or either false claim that the super AI will actually realize very soon the singularity dream and so machines will become more intelligent than humans. So the future is already here. And it is so in many respects, as we see technology already advancing overcoming the weakness and limitation of humans. But in reality, we are probably only at the beginning of this, let's say dive into the future. And clearly we are just starting, realizing, understanding the opportunities but also the huge risks that these technologies brings with them. At the same time expectations from a technology perspective, often a techno-deterministic kind of approach, are not necessarily coupled with the sociopolitical trends. And we have seen an increase in debate on the policy side on our technologies, this kind of technology, have an impact on our society. But we have also seen the need to somehow limit or at least a supervisor the power of the technology giants is just up these days. The you know the adoption and the entry into force of the digital service actually, just to mention one, that we will discuss a bit later on.

**5:07** But there is the need of discussing whether we need to regulate the digital governance landscape, especially because of the, you know, the unexpected consequences that could come from what is known as algorithmic governance. So, there are two main dimensions of concern that have emerged, and are still under debate, and they're clearly not new. One is about the role of the state vis-à-vis the market, the industry, who is actually governing this, you know, the information society, and then the level of individual data protection. So, basically, from a European perspective, we had a debate on the technological sovereignty and the global digital geopolitical landscape that is currently under turmoil, of course, also because of other unfortunate event like the Russian invasion of Ukraine for instance. Now, we have, basically to balance, and we will come back to these at the end, between what Cass Sunstein already 2005 called the precautionary principle and the need of ensuring digital resilience in our society.

**6:18** Now, the scenario, together with a number of experts and colleagues, I crafted back in 2020 looked at these by identifying two main dimensions of impact: one is the digital transformation landscape. So whether we will go into more regulated interventionist approach, or more laissez faire kind of answer on ends of approach. Or on the other side, what is the role of citizen so whether we will have just a passive, let's say approach, as it is, unfortunately, today, in many respects, like for instance, when we just tick the boxes on the GDPR notifications, or on the other side, where we whether citizens will become a real owner of their own data your own data. And basically, we could realize what is called Digital Sovereignty Paradigm. That is, of course, something far to be realized today, were also because of the pandemic in the last few years, the COVID-19 crisis, showed the limitations that we have in our digital systems, and also fear and surveillance on the other side, that came as a priority in many countries. While on the other side, there is exactly this need to include everybody and be more precautionary. But clearly and let's say the ideal scenario that we envision is that of a trusted and open innovation society, where exactly the maker and the shapers of technology actually come together, and realize the positive potential of these technologies for the benefit of society.

**8:14** And this is much different from the current status quo that is, instead, mainly characterized by an empathic condition of the users and closed innovation systems, despite the again the rhetoric of some big tech giants that

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instead think of being, or show to be open, promoting their openness, rather than the closed systems. Now, of course, we all know and futurists know that scenarios are just possible words and words that do not have to be but may yet come to pass. So clearly, the future will be a combination of these technologies, of these scenarios. Also, because of the impact of technology by Napoli, there are many factors that need to be taken into consideration, political and social factors so that we can meet that expectation of citizens, the technological trends that need to be addressed and embodied within the complexity of the socioeconomic systems of our governments and other institutions. That's specifically with regard to the public sector, the need to innovate without necessarily being obsessed by these innovation processes, and at the same time, making sure that this is something real and not just, let's say a maquillage or a mirroring of what is not working offline, online. And basically, this requires and calls for what I called with other colleague, so the need of reframing Public Sector Innovation, so to address these multi dimensional aspects of the digital transformation strategies, in let's say, in a way that is actually considering what are the external factors, the context and making sure that there is a change in the in the process. So I won't go much into detail. But this is actually what is also included in the final report of the study on exploring digital government transformation in the EU that I lead authored for the European Commission Joint Research Centre back in 2020. That includes these scenarios that were actually supporting the policy of the EU on shaping digital government transformation in the EU, and the as the subtitle, say, understanding public sector innovation in a data driven society. So the issue of digital government and data are clearly interconnected. So they are the core of this course, of course, and this comes also alongside the need to ensure responsible AI and digital transformation.

**10:57** This policy brief that was produced exactly during the pandemic with this scenario, also supported the colleagues in the commission and member states in defining what is now, at the moment the latest Declaration on Digital Government. That is, that was adopted by EU member states in December 2020 in Berlin, under the German Presidency of the Council of the European Union, where we can actually see the main, let's say, the main topic, the main issue that are to be agreed, and that were agreed, and that to be respected by EU member states, that are very much not technological. Actually it is how we can use it technology for the respect of fundamental rights and democratic values that are embodied in the EU treaty and the founding value. How we can actually include farther citizens in the digital world. How we can trust and make sure that interaction in the digital government are secure, and exactly how we can guarantee digital sovereignty and interoperability across across the organization. The need of human centric systems, and the novelty of technology, especially in the public sector is also highlighted, as well as the need to go towards a more resilient and sustainable society.

**12:29** In this respect, just to conclude this first part, we see that the future of our society, definitely, will require more you know, to better understand the impact of digital transformation. And so to make sure that citizens actually are in charge of their own data. And clearly for these digital transformation strategies to be effective, it is crucial to reframe, as we say that the public sector systems embracing the complexity of the multiple factors that are underpinning it. And the EU, as this is specifically about the European perspective, has defined a clear policy vision on enhancing digital government for achieving a value based digital society and human centric approach to artificial intelligence, as we will see also later on. And of course, I've only mentioned one of the policy documents that the EU has been consistently developing since the late 90s, beginning of the 20s, on to make sure that openness and innovation are actually at the core of the digital transformation of our society.

**13:43** Now, if we move to what I call the trilemma of governance, “of, with and by” artificial intelligence. So clearly, we see that we need to talk about artificial intelligence. And this was actually the title of a documentary

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that came out in 2020, just during the pandemic, so maybe it passed a bit unobserved, because of course there were more, or the other, more important things to take care of. And, clearly, I would have thought that we need to talk about AI in the public sector, because this was somehow considered not really needed. It is not, you know, the business of the government to make sure that AI is in society, that was the thinking of, at least in the you know, at the beginning of the policy debate on this topic. And the reality AI systems are already an integral part of many, high impact government decisions. There is a mainstream use of automatic decision making systems. So in many very sensitive areas from social benefit entitlements or employment services, and surveillance and security that often are actually based on very poor quality data and algorithmic accuracy.

**15:01** There is also the proliferation of facial recognition systems in public spaces that is actually sometime, in creating unnecessary surveillance and potential risks for human rights respect, and that, of course, non democratic regimes have been farther using, especially during the pandemic. And now are still, of course, reinforcing that. So somehow, the COVID-19 was a high opener in that sense, because clearly, we see that the threats of AI systems actually were farther, let's say, accelerated and we saw that the risk of mishandling the data protection rules, for instance, were higher than before. But the reality this was already the case before COVID-19. And just more pressure was put on to the government, because they had to basically, from in some cases, from day to night or night to day to put, let's say fully on digital services that were just, let's say, as I say before, rhetorically, it claimed to be digital, but they were not digital at all.

**16:22** And that's why in fact, the discussion on the benefits and risks of AI in government is even more actual. And that's why for instance, in 2021, I was asked to by the Digital Futures Society, when I left the European Commission, to write a white paper on the perils and power of algorithms in the public sector. So some of these works also from this research I conducted back then, while I was also the rapporteur of the United Nations on digital government. And we were actually together with the expert group meeting, providing recommendations on how to better capture the phenomenon of artificial intelligence and digital transformation in the post COVID society. As part of the famous e-government survey, global governments survey of UN DESA, the Department of Economic and Social Affairs of the UN.

**17:22** So clearly, we see high potential for artificial intelligence to improve the quality of services because of the capacity of these technologies to process a huge amount of data and support public officials in decision making processes. And so provide tailor made personalized, customized solutions to citizens in very difficult areas and complex social services like social service or employment, as I mentioned before. But on the other side, we see many risks of crystallizing actually, in systems that are already dysfunctional, and at the same time, intensifying the asymmetries that we have in, in power and information and knowledge, and that's penalizing citizens that are already in vulnerable situations. So there are two phenomenon in particular that are there, when it comes to using algorithms and AI systems in different services. We have somehow discrimination by default, because it is true that AI can enable better data collection, help generate knowledge solutions, through predictive analytics and analytics, but also it is often invasive and actually strengthen the prejudice and the biases that we already have in our society. The cases of you know, discriminating women or black people is a clear example.

**18:56** And this is not the only one of course. "Black-boxing effect", because of course, we have to navigate it through the false positives, false negatives. So the algorithms are by definition, prone to error, because they are based on data of a poor quality and that reflect the biases we already have in society. So basically, there are

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paradoxes that we can, that are created by using these systems. And just to even in the terminology, the paradox, we have when we mentioned the blackbox in AI is different from what is, for instance, used in the aviation industry where the black box is instead, the only option, the only possibility to really understand what happened before an accident, for instance. While in this case, entails the fact that we do not know exactly what's happening and how the algorithm was designed, and the data were used, and what kind of data of course. So this, actually is also increasing the risk of mass surveillance as it is, in fact already the case in many countries.

**20:09** And then even in democratic society, like the one in the EU, that combination of COVID-19 on one side, and also the use of technologies in the last couple of years actually put at risk our democratic antibodies, and we've seen several cases in countries that are clearly example of democracy, but that actually showed the limitations of using these technologies. And these are the famous case of SyRi, for instance, the System Risico Indicatie in the Netherlands, and that joint with the system to check the frauds in social benefit entitlements actually shows the limitations of these technologies and the fact that the citizens of poor or immigrant background were actually penalized without need. And this actually brought to the resignation of the government and a new system that actually also demonstrated that the algorithm per se was well optimized. But in reality, the indicators chosen chosen by the policy makers were actually not apt to do to the job. I won't go much into details of the single cases. But we also had an example in Denmark, where they Gladsaxe model, you know, the very noble ambition to trace situations of children at risk. So to avoid that, you know, ghetto situation would be, say, developed in advance society, like the one in Denmark, but actually proved to be discriminatory and biased. And then it was fortunately blocked before being implemented.

**22:02** But, as I said before, there are many cases around the world of use and abuse of facial recognition systems and other biometric remote systems that are actually now under testing and implementation in many countries. So one that we have been included in our repository is the one of the fugitive child robbers in Buenos Aires, that is particularly problematic, exactly, because these technologies are not yet, let's say, so well advanced to ensure that the, you know, the facial recognition of children, I mean, allows to identify the rights children that were in the situation, not to mention the abuse of human rights that these systems also was bringing with it. But similar situations are also in more advanced, let's say, in other countries and cities, from London to Paris, and so forth.

**23:00** Now, another area that has been much debated in how, let's say, artificial intelligence systems can help and in my opinion, it is one of the areas of much let's say, wherever this good technology could help much is in the area of the welfare systems. So the here again, we had the problems in, for instance, in the robotic process automation system used for providing social service support in Sweden, that were again, providing some discriminatory solution. Or the A-Level disgrace, as they call it, of the Ofqual grading algorithmic systems in the United Kingdom during the pandemic, because indeed, by using only an AI enabled system, this was actually creating a situation where children, students, brilliant students, but from disadvantaged backgrounds, were evaluated the less than less brilliant students from more, let's say, advantaged systems. The same is happening in many countries. And we have looked in particular in Austria and Poland, but this is actually already happening in the UK and US, and in many other countries where AI systems are used to provide, you know, unemployed people to support them in their search for employment, and creating, again situations of bias and discrimination.

**24:45** So all this brings me to the you know, the statement of the trilemma of the governance “of, with and by” AI that is basically part of the dilemmas that policymakers have in the digital age. Because on one side they have the



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obligation to protect citizens from potential algorithmic arms, but this, of course, is at odds with the temptation on the other side to increase, and the need actually to, increase efficiency, productivity and enhance quality of digital services by using exactly the same technologies. Now, one side this entails, the need to ensure that there are regulatory systems that do not override the existing governance mechanism and the institutional solution of capacities that need to be of course addressed when introducing artificial intelligence. But there is also the fact that public sector organizations already use quite a lot of artificial intelligence systems. And in this domain, more than in others, it is clear that humans should remain in control, to supervise and take final decisions so that human rights can be safeguarded and AI can be deployed ethically, especially because of the fact that we are dealing with very sensitive situations where this technology that could actually reinforce our capacity as humans, cannot be the one taking the decision for us.

**26:22** But it's true also that the real power, but also the major risk of this technology is actually coming when, you know, the human decision maker somehow surrenders to the superhuman capacity of artificial intelligence. And this can be for the good or for the bad, in the sense that of course, we all dream of systems technologies that actually save us from routine work and boring tasks. And of course, AI have capacity to identify patterns and suggest decisions that we, as humans with our limitations cannot see. But on the other side, we have to make sure that these are ethically, let's say, ethical, and they actually are in line with our ethical value system. So we will be back on these because this is clearly where the true power of this technology lies, but also where we need to make sure that the risks are under control. And of course, if we manage, if policymakers manage to at the same time, say govern these three phenomena, we will actually have found the holy grail of artificial intelligence.

**27:36** Now, what are the lessons learned? Of course, we have to be aware of techno-solutionism. AI is not just a super agent that can do everything. There are no ethical shortcuts. It is not sufficient to put a philosopher in the room to make sure that you know AI systems will behave and be ethical. We actually have to really link the ethical responsible AI systems with the legal regulatory frameworks that we have in place in a democratic society. So we have to adopt a public value perspective, because we cannot just ignore that there are complex challenges, in complex systems in the public sector and in society that we need to address and embrace. We have to be ready to handle disruption, because of course, it's not just, you know, to be fashionable, that we introduce some AI systems, but we actually need to be ready to completely reframe, as I said before a restructuring even not from a cognitive perspective of our public sector system. And then this requires the designing new models of governance, so rethinking how services are delivered, how data are managed, and actually, our entire algorithmic decision systems are implemented.

**28:58** Now, I won't go into details, because I'm sure you will have heard and you already heard or you will, about the AI Act, or, specifically there will be an I'm sure there will be specific classes on these but in reality, now days, we were exactly now we are at the moment where the AI act. So the regulation on artificial intelligence that the European Commission put forward in 2021. And now is coming to a conclusion of the negotiation process with the European Parliament and the Council of the European Union is probably being adopted by the Spanish Presidency of the EU Council before the end of this year. So just to make it let's say in a nutshell, what is important to stay to despite the many, let's say the confusion that sometimes it has been brought into this discussion is that this is a horizontal regulation, so, of course, there will be more specific issues to be addressed at the vertical level in different sectors, but what then most of the actually the practices and use of artificial intelligence where there is no or very low and minimal risks, does not entail any obligations from the users or the

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you know, the organizations using these technologies. But if there are risk, there is the need of making sure that there is a transparent approach. And in this black box, we were mentioned before, and if instead of high risk, these need to be regulated, and there is of course, the requirements for compliance and conformity assessment ex ante, and of course, also with the specific guidelines and standards that are in the making. Now, what is clearly unacceptable or prohibited is what entails unacceptable risk, and this is very much in line with European values, you know, of a democratic system. So, basically discussion here is on remote biometric systems or social credit systems and so forth. Basically all the practices that have the risk to attempt and threaten human rights.

**31:22** Now, in, say, to recap, in summary, clearly we see that the use of AI in the public sector has huge potential, but there are many risks that must be managed by especially improving data quality, but also supervising, auditing algorithms. And there is a need to address the inherent complexity of AI governance exactly, because we need to make sure that if we enable this superhuman capacity of AI, these actually can provide the positive impact. But at the same time, we need to regulate and we need to manage the risk. And we saw some in the AI Act is the first attempt worldwide to govern the use of AI in an ethical, responsible manner. So we can limit the risk of applications that go against the founding values of the European Union.

**32:14** Now, coming to the third part of my presentation, we will look more specifically into the use of AI in public services in the EU. And so here I recall, again, one of the latest report, I've been leader authoring for the Joint Research Centre back in 2020, as part of the AI watch, where I was, indeed, leading the AI in the public sector component. At the time, despite the discussion on the use of AI, the widespread use of AI, we didn't know and still today, we have limited knowledge of what is the real use of AI in the public sector. So we were the first one, definitely the first in Europe to make a mapping of the landscape or the use of AI in public services. And with my team, we identified a number of cases, at the time by discussing directly with all EU member states. And we've identified that we also needed to somehow conceptualize what do we mean by AI-enabled innovation in the public sector, because innovate just for the sake of innovate is not useful. Actually could even be, you know, counterproductive and that's why we actually, to also move away from the discussion of what is AI and the definition of AI, we actually use the functional view of AI, basically looking at the three main characteristics of artificial intelligence systems: perception, reasoning, and action. And we derived a typology from the over 200 cases that we collected in a couple of months in 2020. And where we actually looked not only at the design or the idea, but actually we focused on the adoption and use of application. So to make sure that we don't just have the, the, you know, the political claim of a case, but actually looked at specific technologies and use of technologies in different services.

**34:30** Again, I won't go into details and there has been from my former team at the JRC farther, let's say analysis on this, but at the time, we were already you know, I mean, we already identified some important elements that are still valid today. One is that in reality, there was still a very limited, superficial use of these technologies in the public sector. With most cases actually. Let's say were thought to be innovative but in reality, just making some minor change in the public organization, and very few with the deep transformational of changes in the work practices or making organizational reforms. And again, as we mentioned already, the main challenges are in naturally moving from data analysis to the real adoption of AI systems. And basically many systems were Machine Learning, or I'd say advanced statistical analysis.

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**35:37** Then when it comes to look at the purpose of why we're using this technology, so we still, based on this data set that we had at the time, but this has been further confirmed by further analysis by my former colleagues, is that we have mainly say, support services and trying to engage with citizens through chatbots and matching services that in many cases are still very embryonic or experimental or very basic. We have some cases on detecting fraud like the case of SyRi mentioned before, where several hundreds tried to ensure compliance to regulation with all the limits that we have seen. And also cases of internal support internal management for human resources, procurement and ICT support. Little experience on supporting decision and analysis so predictive analytics to actually gain insights for policymaking, and very little, let's say end to end systems that actually bring to the granting of formal benefits and right to citizens.

**36:47** And, again, what is missing at the moment is the use of the combination of different, let's say, typologies of AI systems. So that actually makes the high potential of these technologies, a set of technology, of these ensemble of technologies. Now, a lot of discussion has also been done on the strategies of Public Strategies of AI. And there's been many analyses of these worldwide. I mean, I've been looking with my team on the specific role of the public sector of AI for the public sector in the European strategies. And we've seen that most EU Member States have by now adopted the specific strategies in the period 2020 until today. And but we also see that I mean, use a metaphor from public policy, we see a lot of sermons but very little carrot and sticks, in the sense that we do see a lot of "yeah, we should do this, we should raise over this, we should build a capacity or learning by doing or defined frameworks, legal and ethical", but there is no real incentive using financial or other means or penalizing systems and organizations that do not comply. So this needs to be further developed. So maybe in the next release of the strategies of the EU countries and worldwide, there will be more, let's say requirements for compliance. And this is also because of the, at least in Europe, of the requirements of the AI Act and the Digital Service Act in the combination, as we will see.

**38:43** So somehow, until now, at least, we have seen a kind of deja vu, as we have seen already in the past, with regard to E-government/e-governance, the Digital Government/Digital governance. So we all look for best practices, and we do a lot of pilots. But we have the syndrome of ever piloting that actually does not allow us to really understand what is really working, and how we can share let's say practices among ourselves. And the need of further reforming our public procurement systems that is very much required if we really want to adopt innovative AI solutions. But also, you know, the understanding and acknowledgement or the multiple roles that the government has in particular, because clearly, we have not yet let's say enough evidence to understand what is the public value created by these technologies in the public sector.

**39:54** So as I said, I left the Joint Research Centre of the European Commission in 2020. The work of my colleagues and colleagues on different commission services, of course, is further advancing on this. And in the coordinated plan on acts of action on AI that the EU and member states adopted, the revised version, in April 2021, there was clearly the need to go beyond pilots, and making sure with a specific section on AI in the public sector, innovation in government is further accelerated. And this requires, as I say, the focus on the procurement side, but also making sure that rather than adequate funding and skills so that the civil servants are empowered to make sure that there is a sustainable adoption of AI based systems.

**40:51** And that's why I will just mentioned briefly the AI4Gov initiative I'm directing, that includes a three areas. One is the master on public services. I mentioned before that is rooted in blended learning and project based work.



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This is specifically focusing among other things on the use of data so by and some work, or the, let's say so called students that we have every year and I mean, we started in 2021. And now we are in the third edition with 40 participants coming from around 30 countries in the world. And we have online lectures and three on site events, where we also meet and discuss with lead experts worldwide. And basically, here we are focusing, we've been focusing, among other things: on procurement, on ethics, and also on the design of the algorithms and AI systems in public service. But also as part of the project on which the master is embedded, we are developing an ecosystem and the Knowledge Hub of excellence on AI for the public sector that is becoming recognized worldwide. So as I mentioned before, this is implemented by the Polytechnic University of Madrid and Politecnico di Milan, with support from the European Commission the EU, under the Connecting Europe Facility Program, also in collaboration with TalTech and FAU, as external power, as additional partners, and also the European Digital Skills and Jobs platform.

**42:35** So this is to mention that I will start here in the summary from the third point. There is a huge need to strengthen the skills of civil servants and policymakers by training the new figures of new profiles or functional specialists in AI, combining skills in design, public management and data science. And this is because, as I said before, there is little evidence of what really works and how we can actually institutionalize AI in public service. And we need to further let's say, focus on data management, in testing of algorithms and predictive models, through sandboxing, for example. And this is also because there is a quite high heterogeneity in the use of AI in the public sector, at least in Europe, but from my experience, worldwide, also on the global level, and we don't have yet a clear understanding of what is the you know, the the impact in terms of public value creation. So we need to further develop these systems in a human centric perspective.

**43:36** And this is exactly the last part of my presentation that will focus on the EU approach to human-centric artificial intelligence. And here again, I won't go into the details. There's been a number of legal and policy acts that have been put forward by the European Union. And if we look only at the last five years, there has been a specific focus on artificial intelligence, on human centric and trusted artificial intelligence through the AI declaration that was signed by all EU member states in April 2018. And the communication of the same here that also, let's say, put emphasis on need to maximize the benefits of AI for Europe. And with a coordinated action plan I mentioned before in the first version, so to make sure that we have a common approach of EU member states, so also to somehow compete in the race on AI where of course the United States, China and other emerging economies are far more advanced. But clearly we have a strong research base in Europe and so the policymakers decided to focus on ethical and responsible AI, trustworthy AI. And this was also the establishment of the high level expert group on AI back in 2018 that also made, you know, suggestions, the guidelines that are now the world then, let's say the inspiration for the AI Act as well as policy and investment recommendations.

**45:25** Now with the Commission found their lie in that interview in the start in 2020, basically, their strategy on shaping Europe's Digital Futures was defined in 2020, with a white paper on AI and the European strategy for data but also very important, although I won't go much into details about the data governance act, the digital service act, and also the digital market act, as I mentioned before, are part of the broader digital governance strategy of the European Union, of which artificial intelligence of course, is an important element. As someone said, "There could be a world before the AI Act, and the world after the AI Act."

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**46:15** So the ambition of the European Union is actually to turn our continent into a global hub for human centric AI. So the AI package that was presented by the commission as a proposal to Member States and the European Parliament on the 21st of April 2021 comprised this proposal of regulation on AI, but also a revised the coordinated plan on AI, on which as I mentioned before, the role of the public sector is acknowledged as a crucial, as a trailblazer, as it is patiently to document for innovation in the public sector. So the combination of the regulation and the coordinated plan with a number of, let's say, initiatives, policy initiatives, investments and cooperation activities, also, the international level, should make sure that citizens can trust what is the offer of AI, so the potential of AI. So making sure that there are these digital transformation strategies, including AI, that are adding a positive impact in the everyday life of citizens, but also strengthening Europe's leading position at the global level.

**47:32** Now, this brings me to the topic of digital sovereignty, I mentioned at the very beginning, that is actually the other side of the coin of global AI governance here. In a paper I wrote in 2020, I was actually during the pandemic, I was raising the question of how we can make sure that digital resilience and collective well being safeguarded, are let's say insured, while we also safeguard liberal democracy, values and individual rights. So basically, from a digital perspective, how we can govern artificial intelligence, why governing “with, and by” artificial intelligence. And definitely this is in line also with the thinking of my good friend Luciano Floridi, who actually in a nice paper on the fight for digital sovereignty, used the metaphor of the investiture controversy that is similar to the situation nowadays. At the time, it was not important whether people were religious or not, but whoever would have won the controversy, I mean, the results of the controversy would have affected everybody in certain parts of the world. And the same is today for the fight for digital sovereignty, will let's say, win the battle I will say, will have an impact on the lives of everybody in the world. But at the same time, we have to recall the precautionary principle I mentioned at the beginning, displayed by the Cass Sunstein in *The Law of Fear* already in 2005 that says that “when negative side effects offset the risks avoided, the precaution can be turned on its head: it is precautionary not to intervene”.

**49:29** So this could be applied to AI in the sense that we don't know the benefits of AI. So better to not use AI, or on the other side, since the AI is actually so good and powerful, we should use them anyway, without taking into consideration the risk. Of course we need to find the balance into this and that's where the European approach to human centric AI is trying to, to put, let's say, a contribution so to make sure that an international shared approach on human centered AI can actually be identified. And this is exactly the objective and the goal of the International Outreach for Human Centric AI. The initiative I have the honor to lead for the European Union that aims to support the AI policy dialogue like the Trade and Technology Council with the US, but also discussion with India, Canada, Japan, and other like minded countries, including Australia, Israel, and Singapore, for instance. And Public Outreach and Technology Diplomacy initiative actually could support, let's say, the creation of a global coalition on human centric AI, and, let's say, a framework that could be agreed at international level for ethics and trust, to enable the growth of AI in accordance with the European and universally recognized values.

**51:01** And recently, together with my colleague, Gry Hasselbalch, as part of the Trade and Technology Council negotiation, there was the definition of 60 terms that should be agreed by starting from the EU and US but possible international level, including human centric AI that we propose as an approach to AI that prioritizes human ethical responsibility, dynamic qualities understanding and meaning and so forth. So the human centric AI systems need to be built on the recognition of a meaningful human technology interaction in which humans

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assume meaningful agency. And so systems need to be designed as tools to serve people with the ultimate aim of increasing human and environmental wellbeing, human respect for the rule of law, human rights, democratic values and sustainable development.

**51:56** Based on these, of course, we cannot ignore that generative AI as now, let's say accelerated the discussion on the potential and risk of AI systems and the use of AI. And as Bill Gates said, in a commentary in March, this year, he was once again after probably what at the beginning of his career was the major excitement he had in terms of technology development, he mentioned that generative AI actually has given rise to the age of AI, that has now finally begun. On the other side that we I mean, we could spend now hours but we don't have time for discussing about the ChatGPT or different version and different generative AI systems. So clearly showed that there is now an interest is immediately of quickly reached capacity, when it was experimented but was also shown by for instance, a study that also one of our students of the AI4Gov master along with other colleagues made experimenting with different queries the limits of the conventional conversational AI systems of ChatGPT. There seems to be actually imperfect by design.

**53:30** Clearly a lot needs to be done to make sure that AI can actually indeed, help augmenting human capacity as it can. But only if ethically mastered. In this sense, I just want to conclude with, you know, recalling, and recently in June, I had the pleasure to meet Sir Tim Berners-Lee, in an event. Basically AI and generative could actually help in realizing what is the quintessential innovator's dream and clearly Tim Berners-Lee as the creator of the world wide web is an example of a quintessential innovator dreamer. So a kind of Back to the Future return to the utopian origins of the internet so that we could promote a worldwide the building of an open, democratic and sustainable digital society. So the open question I leave you with is, if the future of AI should be found in the origins of the internet. So a very much, let's say hot discussion now with the summit of the future of the UN next year, that will also debate among other things, the need for a digital governance framework for our society.

**54:49** So to conclude, realizing European vision for human centric AI could definitely enable you to narrow the EU to narrow the gap between the leading countries in the world. But as the Secretary of State of the Spanish government on Digital Transformation and AI says in the last year event at the AI4Gov Masters, the European Union approach is not just a third way between US and China, but is actually could be the only way forward to develop truly ethical and human centric AI systems. Of course, the explosion of generative AI systems must be taken into account so we can anticipate risk but also exploit the huge opportunities they offer for increasing the capabilities of individuals, organizations and society. And the issue of digital sovereignty that I mentioned several times today, it should be addressed jointly as an opportunity to increase the strategic autonomy of EU member states, but not only, also of like minded partner countries, so to address the together the global challenges of digital governance and technology diplomacy in the for the future of our society. So with this, I thank you. And, of course, I'll be glad to be in touch with you and make sure that we all together can achieve the quintessential innovator's dream of a human centric artificial intelligence at the global level. Thank you.